Load Balancers - OSI Layers and Packet Forwarding

A single server is a single point of failure and a finite bottleneck. In distributed architectures, load balancers act as the traffic directors, distributing incoming network requests across a pool of resources to ensure high availability, fault tolerance, and horizontal scalability. However, not all load balancers are created equal. They operate at different layers of the OSI networking model—specifically Layers 3, 4, and 7—which dictates how much of the traffic they can see and how they make routing decisions. L3 load balancing routes traffic based purely on IP addresses, often utilizing network routing protocols like BGP and techniques like ECMP (Equal-Cost Multi-Path) or Anycast. In most architectures, L3 routing sits at the cloud provider's edge infrastructure to absorb massive scale, rather than acting as a user-configured application gateway. Operating at the TCP and UDP level, an L4 load balancer distributes traffic based entirely on the source and destination IP addresses and ports. **Mechanism:** It moves packets mechanically without parsing them. It does not inspect the payload, allowing it to forward traffic with very low per-connection CPU overhead and latency. L7 load balancers operate at the HTTP, HTTPS, gRPC, and WebSocket level. **Mechanism:** Unlike L4, an L7 proxy actively terminates the client connection. It opens the packet to inspect the application payload (reading URL paths, host headers, and cookies), makes an intelligent routing decision, and establishes a second, separate connection to the backend server. Historically, load balancing relied on specialized **hardware appliances** (like F5 or Citrix) deployed in on-premises data centers. These physical devices provide massive throughput via custom ASICs, but they are expensive, inflexible, and difficult to scale dynamically. Conversely, **software load balancers** (such as NGINX, HAProxy, and Envoy) run on commodity compute instances or as fully managed cloud services. They are the backbone of cloud-native architecture, prioritizing agility, programmatic configuration, and dynamic autoscaling over dedicated hardware acceleration.

Tradeoffs and Use Cases

Choosing between L4 and L7 is a direct tradeoff between pure throughput and routing intelligence. In modern microservices, architects often layer them: an L4 proxy at the edge handles connection volume, passing traffic to internal L7 proxies for precise routing.

Layer	Primary Use Cases	Tradeoffs
L4	Maximum throughput, connection-heavy workloads, non-HTTP protocols, or when TLS must be terminated on the backend.	Blind to application context. Cannot perform path-based routing, manage cookies, or enforce web application firewalls (WAF).
L7	Microservices API routing, central TLS management, path/host-based traffic shaping, and complex web applications.	Higher latency and CPU overhead due to connection termination and payload parsing.

Both major hyperscalers provide managed, software-defined load balancers, but their architectural philosophies and product lines differ. AWS separates its load balancers into distinct, purpose-built products: **Network Load Balancer (NLB):** The L4 solution. It supports TCP, UDP, and TLS traffic, routing via a flow hash algorithm. It provides extreme performance

and offers static IP addresses. **Application Load Balancer (ALB):** The L7 proxy. It is designed for HTTP/HTTPS/gRPC routing based on application-level data and integrates deeply with AWS container services. **Gateway Load Balancer (GLB):** Operates across L3 and L4 to seamlessly route traffic through fleets of third-party network and security appliances. Google Cloud utilizes a software-defined approach built heavily on its global network backbone: **Application Load Balancers:** Google Cloud's L7 proxy solution. When deployed globally over the Premium Tier, it uses a single Anycast IP to terminate connections worldwide, directing traffic to the closest healthy backend. **Network Load Balancers:** Google Cloud’s L4 offerings are uniquely split into two architectural models: **Proxy Network Load Balancers:** Layer 4 reverse proxies that terminate TCP sessions at the load balancing layer before forwarding them. **Passthrough Network Load Balancers:** Layer 4 load balancers that do not proxy connections. They forward the original client packets with unchanged source IPs directly to regional backends.